

AGU 2025 abstract: Multi-instrument Synoptic Characterization of Deep Convective Events and Associated Gravity Wave Responses Over CONUS Using AWE, AIRS, TEC, and LLITED

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Abstract NASA's Atmospheric Waves Experiment (AWE), launched aboard the International Space Station on 9 November 2023, provides high-resolution imaging of mesospheric hydroxyl (OH) airglow emissions near 87 km altitude. Mounted on the ISS (~400 km altitude, 51.6° inclination), AWE captures geo-referenced images of gravity waves (GWs) with ~1.75 km horizontal resolution and ~10s cadence during nighttime orbital segments. The instrument's swath width spans approximately 300 km in the cross-track direction, with observations extending from ~±51.6° latitude every ~90 minutes. This sampling enables near-global coverage over time and facilitates multi-scale, multi-local-time characterization of atmospheric gravity wave (AGW) activity in the mesosphere.

We focus on the May 26, 2024, tornado outbreak over Texas. This high-impact convective event produced widespread damage and served as a strong source of vertically propagating AGWs as seen by AWE, which captured signatures of waves generated by the storm before the development of the tornado. NEXRAD radar reflectivity was used to identify convective storm cores and track their evolution. AWE observations were compared to GNSS-derived total electron content (TEC) from the System for Rapid Analysis of Ionospheric Dynamics (S-RAID; Inchin et al., 2023), stratospheric brightness temperature perturbations from AIRS, and in situ thermospheric density measurements from the LLITED mission. A custom analysis pipeline—incorporating keogram generation, and Fourier and wavelet spectral methods—was used to trace wave signatures from their convective sources through the stratosphere (AIRS), mesosphere (AWE), and ionosphere (vTEC).

Wavefront analyses reveal concentric AGW patterns across all imaging instruments, with phase speeds, horizontal wavelengths, and propagation directions consistent with a broad spectrum of upward-propagating convectively generated gravity waves. At stratospheric and mesospheric altitudes observed by AIRS and AWE, respectively, horizontal wavelengths range from ~40 to 70 km. In comparison, GNSS ionospheric pierce point (IPP) data indicate MSTID structures with horizontal scales near ~400 km. Both AIRS and AWE brightness perturbations exhibit strong spatial and time-of-arrival correlation with the convective sources and TEC-based MSTIDs and acoustic signatures. In situ measurements from LLITED further confirm thermospheric responses to these events. These findings underscore the value of coordinated, multi-layer observations for resolving AGW source attribution, vertical coupling dynamics, and broader MLT impacts and their climatological implications.
